Ezairo® 7150 SL Hybrid Demonstrator Board User's Guide

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ON Semiconductor®

Ezairo 7150 SL Hybrid Demonstrator Board User's Guide

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CHAPTER 1
Introduction

1.1 Purpose

This document provides detailed information for the configuration and use of the Ezairo® 7150 SL hybrid demonstrator board. The hybrid demonstrator board consists of two printed circuit boards: a motherboard and daughterboard that are designed for use with the software tools included with the Evaluation and Development Kit (EDK) to evaluate the performance and capabilities of the Ezairo 7150 SL hybrid.

The Ezairo 7150 SL hybrid is a System-in-Package (SiP) that includes the following:

- ON Semiconductor Ezairo 7100 digital signal processor (DSP)
- Nordic Semiconductor nRF51822 2.4 GHz radio IC
- ON Semiconductor 2 Mb EEPROM (EA2M) for audiology

1.2 Intended Audience

This manual is for engineers, developers, and anyone else who requires technical information for the hybrid demonstrator board, or who wants to use the board for evaluation and development.

1.3 Conventions

The following conventions are used in this manual to signify particular types of information:

- Component and pin names, as well as file and path names are in a `monospace font`.
- Variable path names (where you fill in suitable information) and characters that you type are in a **`bold monospace font`**.
- Default jumper configurations and items you see on the screen are in a **bold typeface**.

1.4 Manual Organization

The *Ezairo 7150 SL Hybrid Demonstrator Board User's Guide* contains the following chapters and appendices:

- **Chapter 1: Introduction**, describes the purpose and organization of this manual, the intended audience, and provides a list of recommended reading for more information.
- **Chapter 2: Overview**, provides an overview of the hybrid demonstrator board for Ezairo 7150 SL.
- **Chapter 3: Getting Started**, explains how to use the hybrid demonstrator board, including powering it up, communicating with it, then programming the nRF51288 radio IC and the Ezairo 7100 chip, and connecting the audio input/output.
- **Chapter 4: Hybrid Demonstrator Board**, provides the details of the hybrid demonstrator board. The chapter is divided into the following topics:
  - The setup
  - The design
  - Power supply
  - Digital input/output
  - I^2C
  - CS44
  - Audio interfaces
  - Board configuration

- **Appendix A: Connectors**, provides a complete list of the connectors and jumpers on the hybrid demonstrator board.
- **Appendix B: Motherboard Schematics**, contains the motherboard schematics for the hybrid demonstrator board.
- **Appendix C: Daughterboard Schematics**, contains the daughterboard schematics for the hybrid demonstrator board.

1.5 Further Reading

For more information, refer to the following documents:

- *Ezairo 7150 SL BTE Hearing Aid Reference Design Firmware Manual*
- *Ezairo 7150 SL Remote Dongle Reference Design Firmware Manual*
- *Promira Serial Platform Quick Start Guide*, available at totalphase.com
- *Communication Accelerator Adaptor Manual*, which is installed with the Communication Accelerator Adaptor (CAA) software
- *Integrated Development Environment User's Guide for Ezairo 7100*
- *Introduction to Ezairo 7100 Programming*
- *CFX DSP Architecture Manual*
- *Hardware Reference Manual for Ezairo 7100*
- *HEAR Configurable Accelerator Reference Manual*
- *Firmware Reference Manual for Ezairo 7100*
- *Ezairo 7150 SL Firmware Reference Manual*
- *SK4 ELF Toolchain Reference for Ezairo 7100*
- Ezairo 7150 SL Datasheet
- EA2M Datasheet (Available at www.onsemi.com)

CHAPTER 2

Overview

2.1 Introduction

The Ezairo 7150 SL hybrid demonstrator board consists of a motherboard and daughterboard used for evaluating the Ezairo 7150 SL hybrid. The motherboard provides access to all input and output connections via 0.1” standard headers. One 6-pin DIN socket (J2) is available on the motherboard to connect to the Total Phase™ Promira™ Serial Platform or Communication Accelerator Adaptor (CAA) that provides communication to the board from a host PC. These connectors can also be used for connecting to other Ezairo 5900, Ezairo 6200, or Ezairo 7100 platforms. For more information about the Promira, see the *Promira Serial Platform Quick Start Guide*. For more information about the CAA, see the *Communication Accelerator Adaptor Manual*, which is installed with the CAA software.

The daughterboard connects the Ezairo 7150 SL chip to the motherboard, and can be powered independently of the motherboard. The daughterboard was designed for evaluating Ezairo 7150 SL as a smaller form factor, and it is used for key measurements such as power consumption.

2.2 Hybrid Demonstrator Board Features

The hybrid demonstrator board offers unique advantages, enabling developers to evaluate the performance and capabilities of the Ezairo 7150 SL hybrid in addition to developing, demonstrating and debugging applications.

NOTE: The motherboard is required for programming the Ezairo 7150 SL chip and requires the daughterboard. The daughterboard can be used by itself.

The motherboard includes:

- Support for manually powering RFVDD, VDDO2, and VDDO3 when the appropriate resistors on the daughterboard are removed
- Support for various VBAT powering options
- Two buttons for application usage purposes
- Standard audio connectors (0.1” headers)
- Access to all Ezairo 7150 SL digital input/output signals via standard 0.1” headers
- Easy interface to the Promira Serial Platform or CAA for communication from a host PC to Ezairo 7150 SL
- Provision for adding a CS44 connector

CHAPTER 3
Getting Started

3.1 Introduction

This chapter is about setting up the hybrid demonstrator board for use. You have to provide it with power and connect it to your computer. If you want to program the nRF51822 radio IC and the Ezairo 7100 on the board, you need some software development tools. Finally, if you would like to evaluate the volume control, this chapter describes connecting to the audio input/output.

You need a programming box to connect the hybrid demonstrator board to a computer. Supported programmers are:

- Promira Serial Interface from Total Phase, Inc., driver version 1.3.6[1]
- Communication Accelerator Adaptor (CAA), driver version 1.3
- HI-PRO, driver version 2.0.0.4

3.2 Powering the Ezairo 7150 SL Motherboard

1. Connect the J-link Debugger to the header J1 on the motherboard. Note that if communications do not work, turn the connector 180 degrees and try the other orientation.
2. Connect the programmer to the header J2 on the motherboard.
3. Short the header VBAT-I.
4. Short the top row (pins 1 and 2) of PSU-SEL. This will cause the system to be powered by a battery connected to J3. Additional powering options are noted in Section 3.2.1, “Selecting the Power Supply”.
5. Switch SW3 to **ON**.

1. The Promira Serial Platform is included in the Ezairo 7100 Evaluation and Development Kit (EDK).

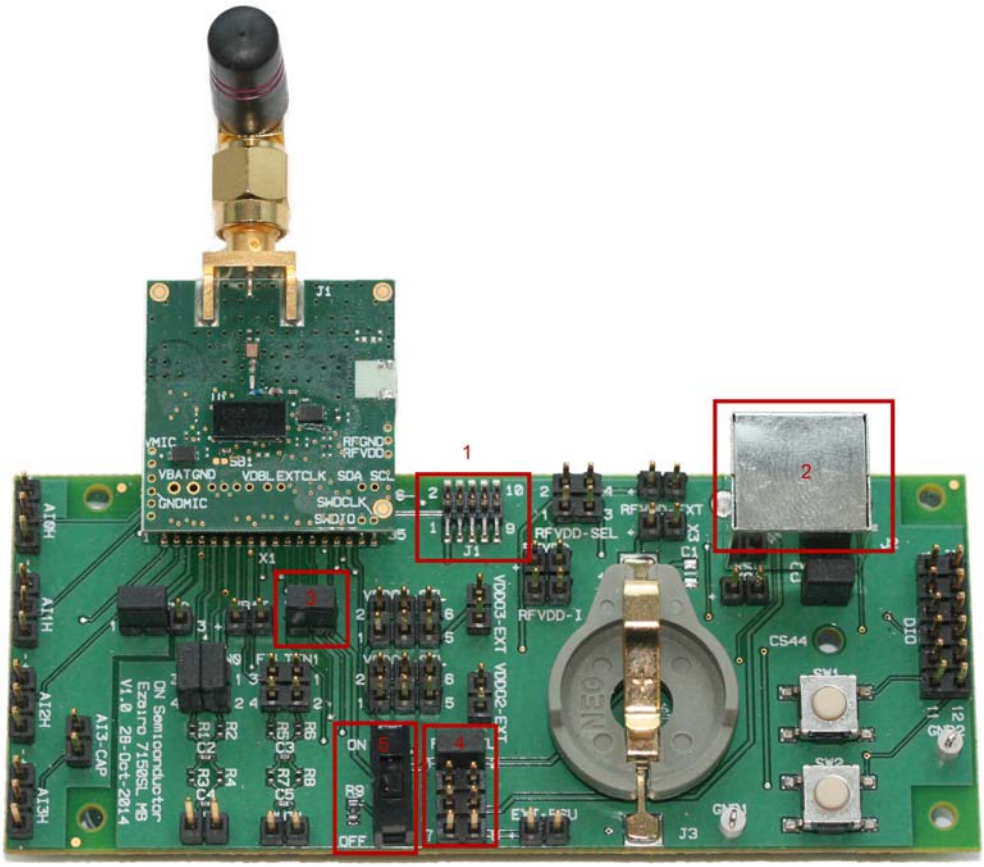


Figure 1. Ezairo 7150 SL Motherboard

3.2.1 Selecting the Power Supply

To change the power supply for Ezairo 7150 SL, a different row of the header, **PSU-SEL** must be shorted.

Table 1. Ezairo 7150 SL Motherboard Powering Configuration Options

Pin Numbers	Power Supply	Additional Connections
1 and 2	Battery	A battery must be placed in the battery clip J3
5 and 6	Programmer	Programmer must be connected
7 and 8	External Power Supply	An external power supply must be connected to EXT-PSU (positive polarity is marked by the '+' sign)

3.3 PROGRAMMING NRF51822

To program nRF51822, download and install a copy of nRFgo Studio from the Nordic Semiconductor website. The program is available for both 64 and 32-bit operating systems.

The sequence of operations below is essential for proper operation of the device. The radio must be erased first, followed by the programming of the SoftDevice, and finally, programming the application.

1. Open **nRFgo Studio** (use version 1.17.1 or higher).
2. Under the **Device Manager** window, click on **nRF51 Programming**.
3. Erase the device by clicking **Erase all**.
4. Select the **Browse** option. If you are programming the hearing aid reference design, locate the file *s110_nrf51*.hex*. If you are programming the Remote Dongle reference design, locate the file *s120_nrf51*.hex*.
5. Click on the **Program** button.
6. Click on the **Verify** button.
7. Select the **Program Application** tab and select the **Browse** option. Locate one of the following files:
   - For the hearing aid reference design, *Ezairo7150SL_HearingAid_Sample_`datarate`.hex*
   - For the Remote Dongle reference design, *Ezairo7150SL_RemoteDongle_Sample_`datarate`.hex*

   where `datarate` is *ldr* (Low Data Rate) for 48 kbps, or *hdr* (High Data Rate) for 96 kbps.

   (`*` indicates the version of the file). Select the P**rogram** button.

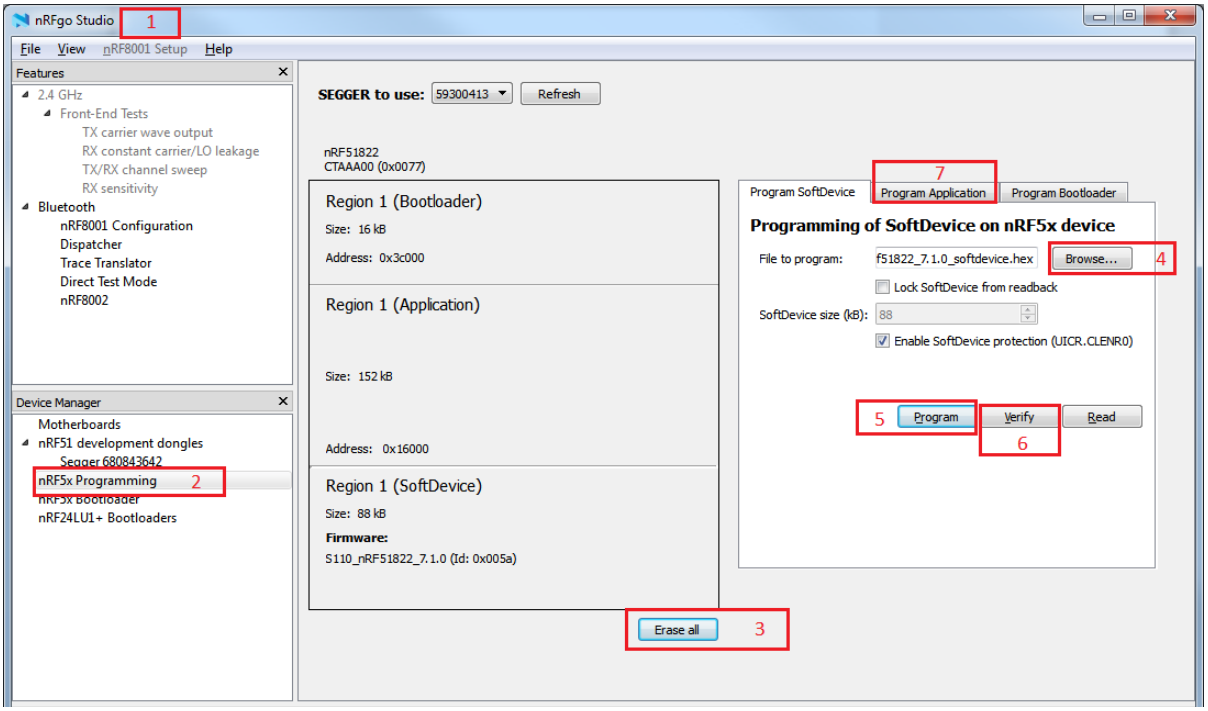


Figure 2. Interfaces for Erasing the Radio and Programming the Device and Application

3.4 Configuring the Programmer

To program Ezairo 7150 SL, the programmer must be configured properly, as follows:

1. Open the Ezairo 7100 Development tool IDE.
2. Select **Tools** on the toolbar at the top of the IDE.
3. Select **Unlock Device...** from the drop box tab.
4. Under **Communication Configuration**, select **Configure**.
5. In the new window, click the **Add** option.

6. If you have a CAA, match the settings found in Figure 3. Settings for Promira and HI-PRO are provided below.

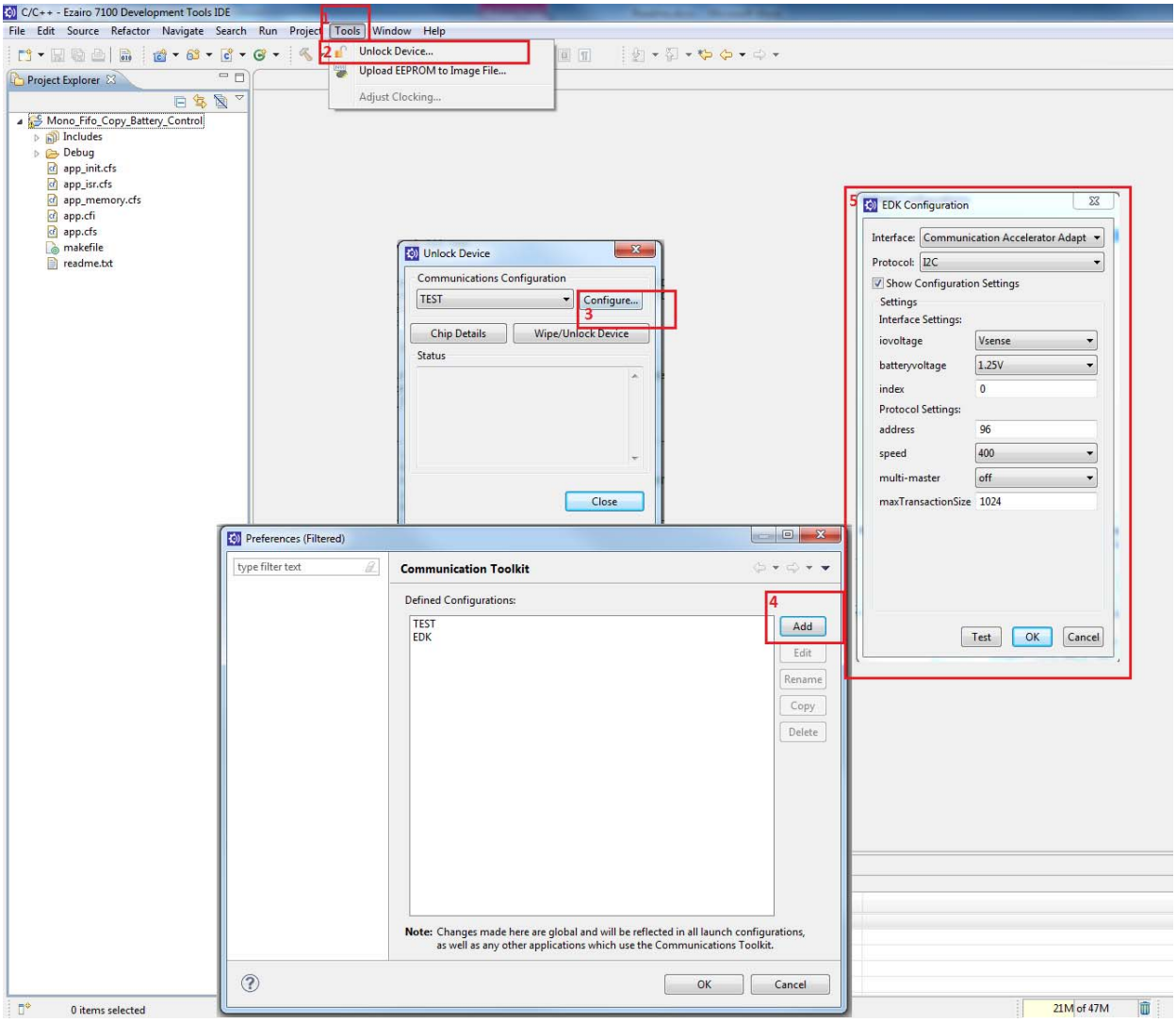


Figure 3. Interfaces for Configuring the Programmer

To configure Promira, set the fields as described in Table 2.

Table 2. Promira Communication Configuration

Field	Setting
Interface	Promira
Protocol	I2C
Show Configuration Settings	checked

Table 2. Promira Communication Configuration (Continued)

Field	Setting
serialNumber	Enter the serial number on the bottom of the Promira, and this configuration will only use that programmer. Alternatively, leave this field blank and the configuration will use the first available Promira that it finds.
targetpower	on
targetpowerlevel	1.25V
address	96
speed	400
pullups	enabled
maxTransactionSize	1024

To configure HI-PRO, set the fields as described in Table 3.

NOTE: To use a HI-PRO, ensure that you have downloaded the drivers and put their location in the computer path.

Table 3. HI-PRO Communication Configuration

Field	Setting
Interface	HI-PRO
Protocol	I2C
Show Configuration Settings	checked
ear	Left
port	default
iovoltage	1.35V
hiprodriverversion	402
address	96
speed	400

3.5 Loading the Project into the IDE

To use the tools provided by the Ezairo 7100 Integrated Development Environment (IDE), you must load the project into the IDE:

1. Open the Ezairo 7100 Development tool IDE and right click the area under Project Explorer. Select **Import** from the drop-down box.
2. Under the **General** tab, select **Existing Project from Workspace**.
3. For root directory, select the project folder **Ezairo 7100**.
4. If you are using the hearing aid reference design, use the `Demo_HearingAid` project. If you are using the Remote Dongle reference design, use the `Demo_RemoteDongle` project.
5. Click on **Finish**.

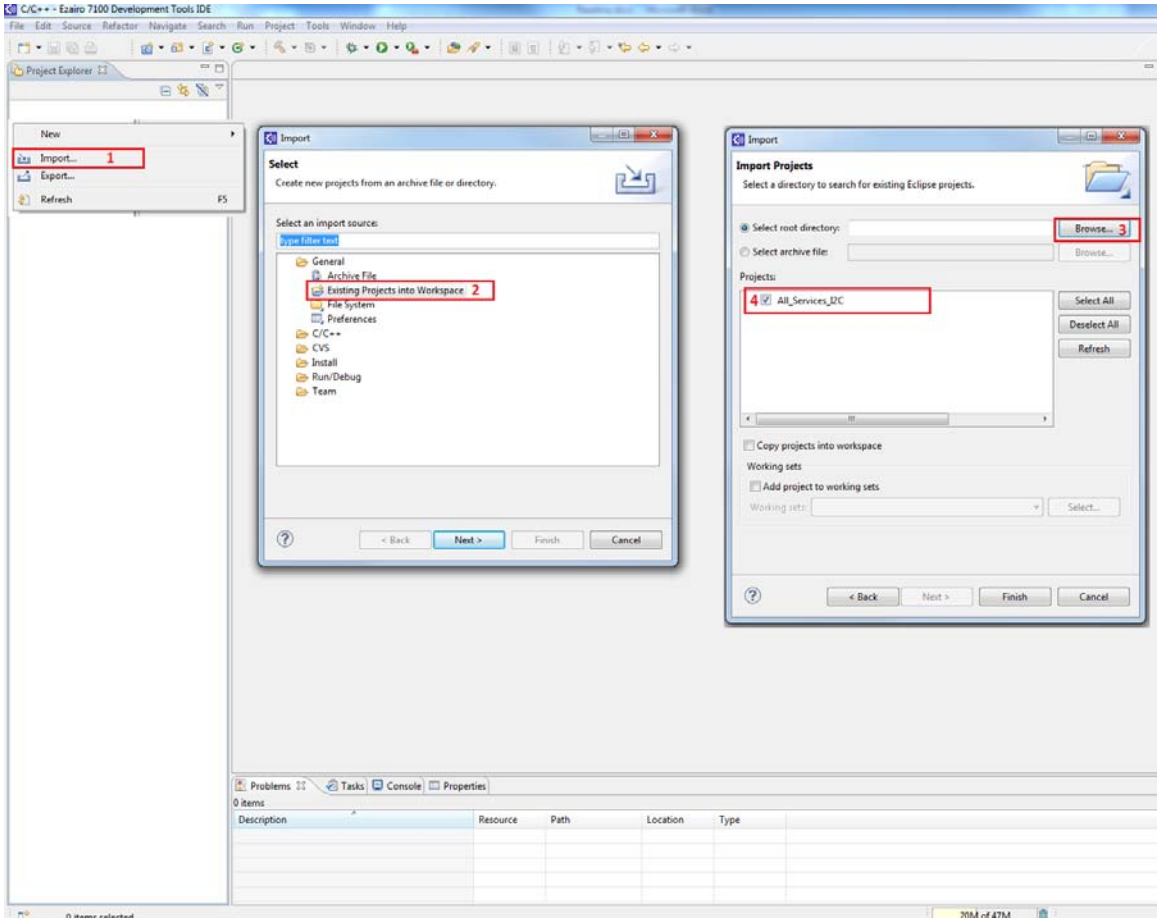


Figure 4. Loading the Project

3.6 PROGRAMMING EZAIRO 7100

The demo projects contain different NVI files to use depending on the data rate you want, and whether you are working with the hearing aid reference design or the Remote Dongle reference design:

- For the hearing aid reference design:
  - Use *Demo_HearingAid_ldr.nvi* for a 48 kbps data rate (Low Data Rate).
  - Use *Demo_HearingAid_hdr.nvi* for a 96 kbps data rate (High Data Rate).

- For the Remote Dongle reference design:
  - Use *Demo_RemoteDongle_ldr.nvi* for a 48 kbps data rate (Low Data Rate).
  - Use *Demo_RemoteDongle_hdr.nvi* for a 96 kbps data rate (High Data Rate).

Once the project is in the IDE, you can use that tool to load it into the EEPROM on the chip.

1. Right click the project and select **Build Project**.
2. Highlight the **Run As** option, followed by the **Downloaded SK4 NVI** option.

3. Choose the relevant file to launch depending on the data rate: *Demo_HearingAid_ldr.nvi* for 48 kbps or *Demo_HearingAid_hdr.nvi* for 96 kbps.
4. Check the **All_Services_I2C** box.
5. Click on **Run**.

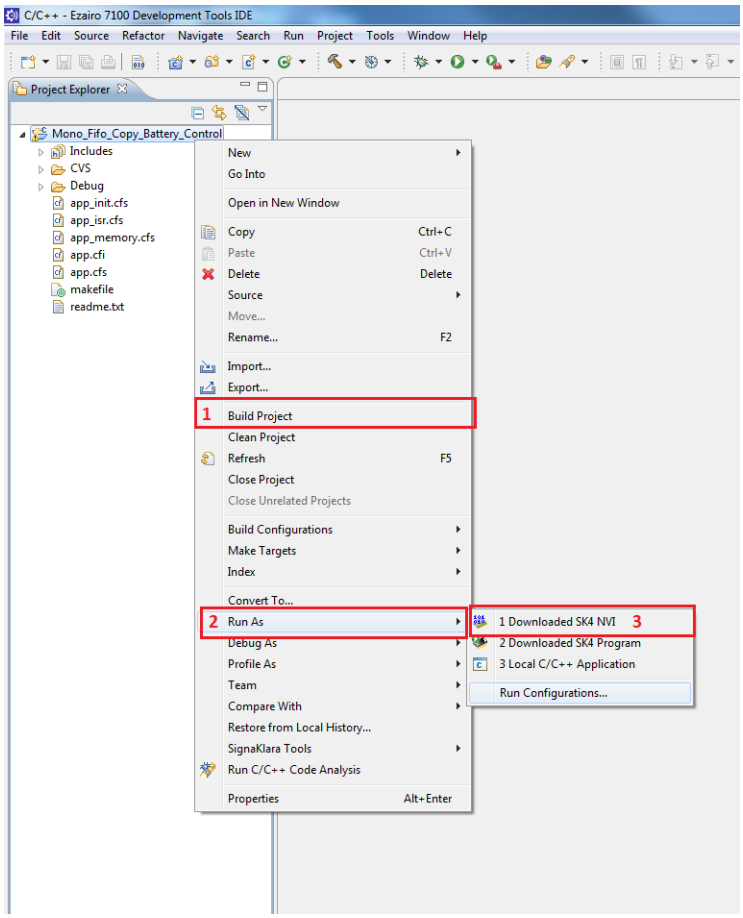


Figure 5. Using the IDE to Program Ezairo 7100

3.7 Connecting Audio Input/ Output (I/O) to the Ezairo 7150 SL Motherboard

To easily evaluate the volume control featured on the Android demo application, you have to connect an audio source to the `AI1H` input header.

Connect a microphone or line level input to the `AI1H` jack as shown in Figure 6 below:

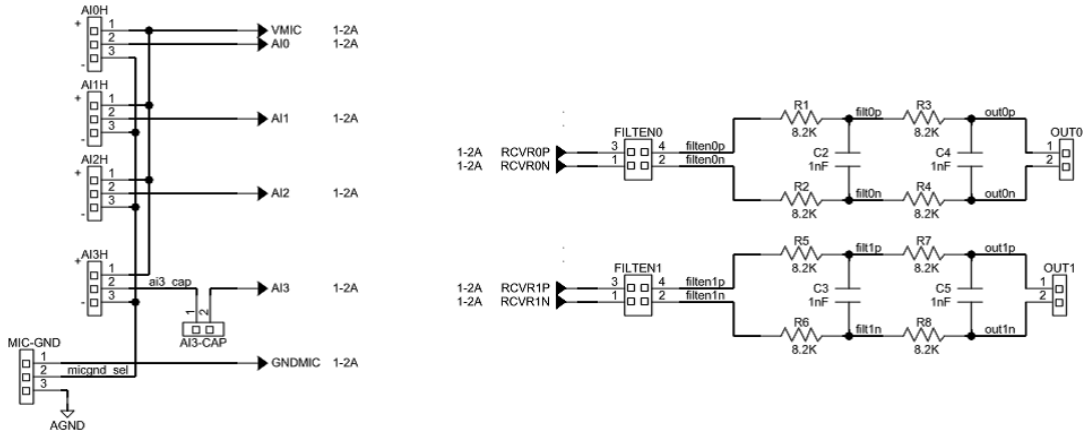


Figure 6. Audio Input/Output Schematic for the Motherboard

The input 3 position header is oriented with the center pin connected to the audio input.

Connect a hearing aid receiver to pins 1 and 3 of the `FILTEN0` header, or jumper this header and connect a line level monitoring device to the `OUT0` header.

3.8 Using the Hybrid Demonstrator Board as a Hearing Aid or Remote Dongle

When using the board as a hearing aid, `SW2` on the motherboard acts as the button on the hearing aid. When using the board as a Remote Dongle, the buttons on the motherboard map to the Remote Dongle as follows:

- `SW1` = UP button
- `SW2` = DOWN button

If you are using the board as a Remote Dongle, you also have to change the *shared_defines.h* file to select the appropriate hardware platform. Specify the `HARDWARE_PLATFORM` define as follows:

```
/* Hardware configuration - Select between Remote Dongle and Demo Board */
#define HW_REMOTE_DONGLE       0
#define HW_EDK                 1
#define HARDWARE_PLATFORM      HW_REMOTE_DONGLE
```

For more information about using these buttons, see the *Ezairo 7150 SL BTE Hearing Aid Reference Design Firmware Manual* or the *Ezairo 7150 SL Remote Dongle Reference Design Firmware Manual*.

CHAPTER 4
Hybrid Demonstrator Board

4.1 Demonstrator Board Setup

This section is an overview of how to configure the hybrid demonstrator board. Figure 7 represents an overview of the board setup. Communication to the Ezairo 7150 SL hybrid requires both a daughterboard and a motherboard.

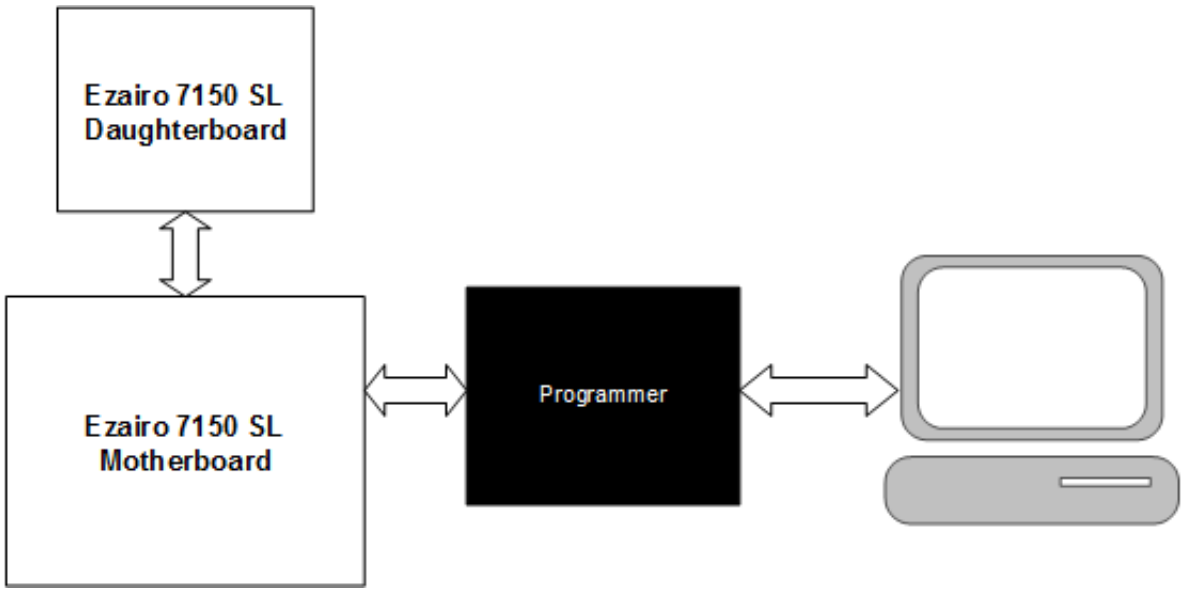


Figure 7. Hybrid Demonstrator Board Setup

The Ezairo 7150 SL hybrid is packaged in a 49-pin ball grid array (BGA) package that is used on the hybrid demonstrator board.

The Ezairo 7150 SL hybrid demonstrator board can be powered from different sources via the motherboard:

1. A programmer (see Section 3.1, “Introduction” on page 7 for a list of supported programmers)
2. An external power source
3. A hearing aid battery

4.2 Development Board Design

The following sections detail the various sub-circuits of the Ezairo 7150 SL hybrid demonstrator board; both boards are outlined. The block diagram in Figure 8 shows the locations of the various circuit sections for the motherboard.

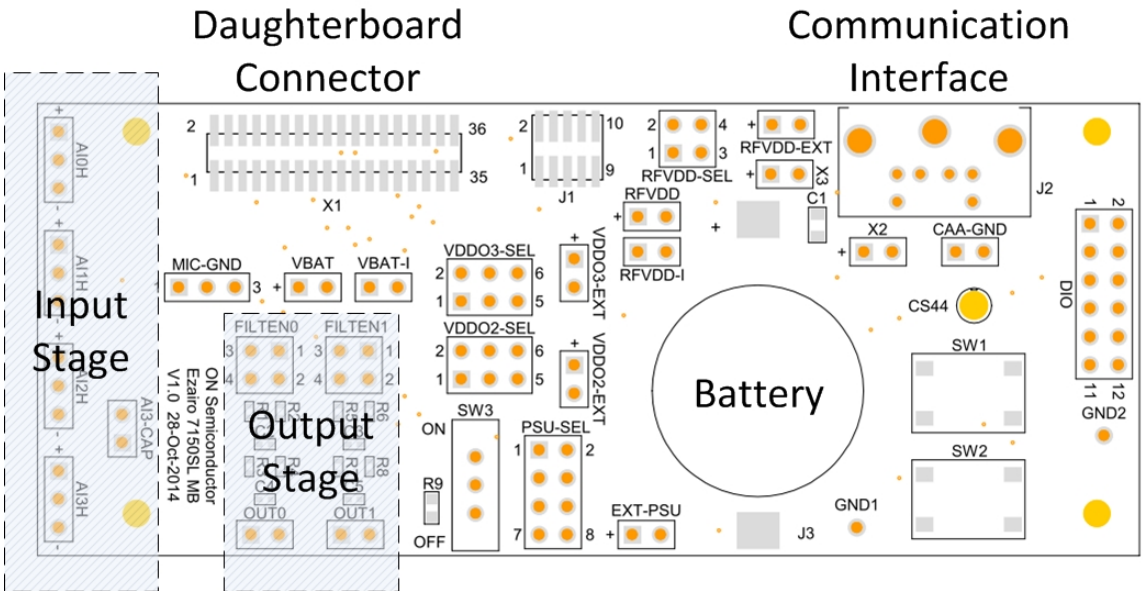


Figure 8. Motherboard Circuit Location Block Diagram

A block diagram showing all of the daughterboard circuit sections is shown in Figure 9.

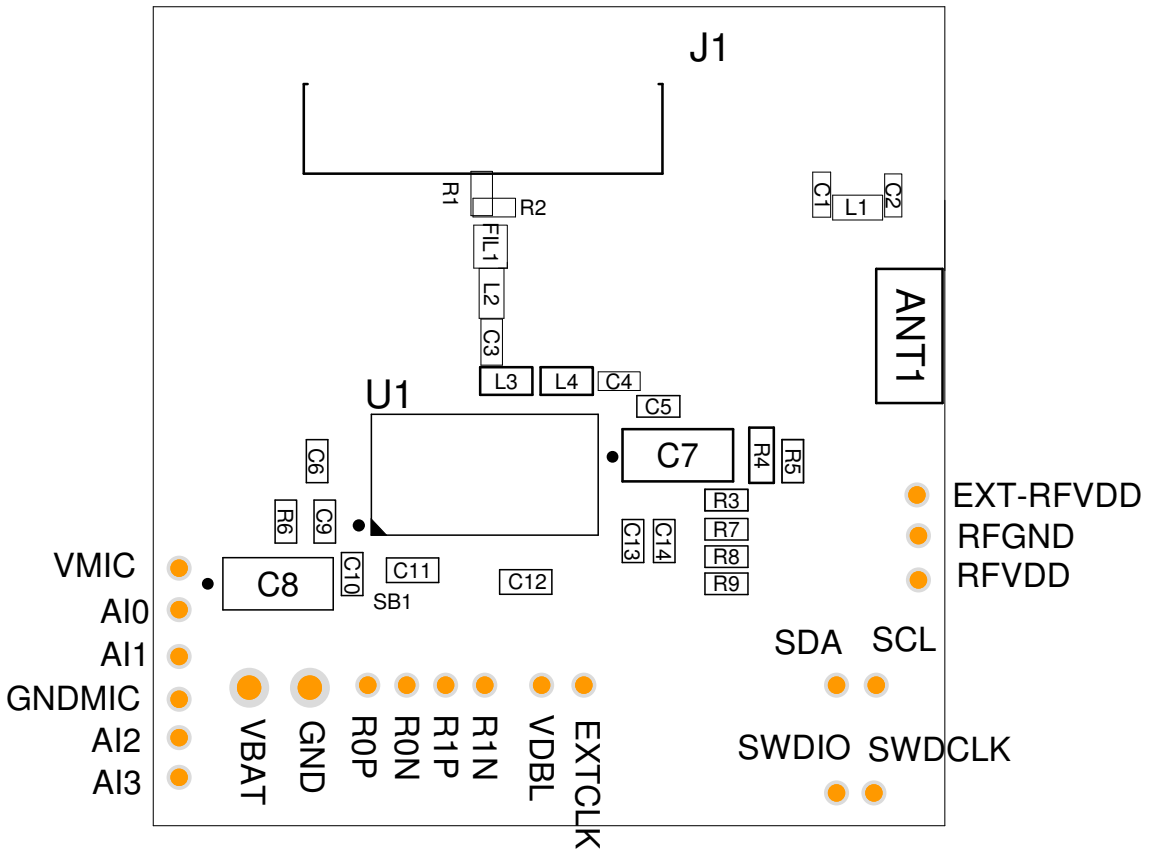


Figure 9. Daughterboard Circuit Location Block Diagram

The motherboard connects directly to the daughterboard via a 36-pin header (X1). This is shown in Figure 4.

For development purposes, we recommend using the following off-the-shelf SMA antenna: ANT-2.4-CW-RAH-SMA-ND from Linx Technologies.

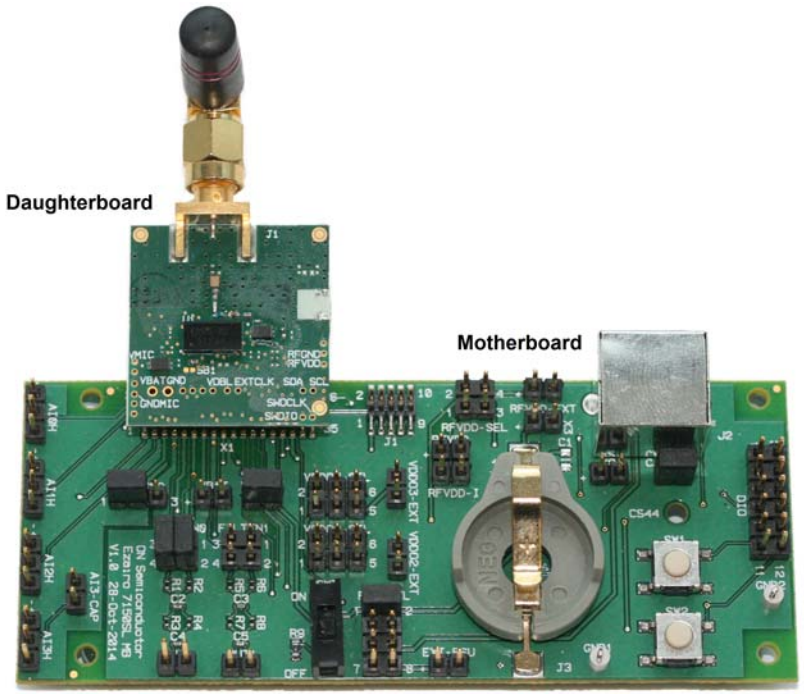


Figure 10. Motherboard/Daughterboard Connection

4.3 Power Supply

The Evaluation and Development Platform can be powered using the motherboard and daughterboard connected together, or by using the daughterboard alone. The main power supply pin for Ezairo 7150 SL is VBAT. When using the motherboard with the daughterboard, the Ezairo 7150 SL hybrid can be powered in the following ways:

- Via the Promira Serial Platform, connected by the 6-pin DIN header which accesses connector `J2`. Alternatively, use the CAA or HI-PRO.
- Via an external supply connected to the `EXT-PSU` header
- Via a hearing aid battery (size 312, 13 or 675) inserted into the 16 mm battery holder (`J3`) located on the motherboard

The `VBAT-I` header located on the motherboard is provided for measuring the current consumption of Ezairo 7150 SL. For normal operation, short the `VBAT-I` header. When using an external supply, ensure the Ezairo 7150 SL recommended voltage level is not exceeded; refer to Table 4 for minimum and maximum voltages.

Table 4. VBAT Minimum and Maximum Voltages

Minimum (V)	Maximum (V)
1.05	2.00

The `PSU-SEL` switch allows you to select between any of the powering options noted above by connecting a two-pin jumper. This is described in more detail in Table 5. Note that any pins that are not specified as being shorted are intended to be left open.

Table 5. Power Supply Selection

Power Source	PSU-SEL	Additional Connections
Programmer	Short Pin 5 and 6	Adaptor must be connected
Battery	Short Pin 1 and 2	A battery must be placed in the battery clip J3
External Supply	Short Pin 7 and 8	An external power supply must be connected to EXT-PSU (positive polarity is marked by the '+' sign)

When using the daughterboard standalone, you must use external power via the VBAT and GND pins shown in Figure 9 on page 17.

Ezairo 7150 SL provides access to 10 digital I/Os that are mapped into two power domains: VDDIO1 and VDDIO3.

The headers VDDO2-SEL and VDDO3-SEL (refer to Table 6) configure the I/O voltages for each of these power domains to either VBAT, VDBL (a regulated supply from Ezairo 7100), or an external supply. The VDDO1 I/O voltage is internally connected within the hybrid to VDBL and is used to power the EA2M that is inside the hybrid. The VDDO2 I/O voltage powers the I/O signals connected between Ezairo 7100 and nRF51822 inside the hybrid. The VDDO3 I/O voltage is used by Ezairo 7150 SL for its debug port interface reset pin.

Table 6. I/O Voltage Configurations

Header	VDBL	VBAT	External Supply
VDDO2-SEL	Short pins 1 to 2	Short pins 3 to 4	Short pins 5 to 6
VDDO3-SEL	Short pins 1 to 2	Short pins 3 to 4	Short pins 5 to 6

4.4 Digital Input/Output (DIO)

The motherboard provides access to all of the DIOs exposed on the Ezairo 7150 SL hybrid on the DIO header. The header also provides two DGND pins for prototyping purposes, as well as two buttons tied to DIO24 and DIO29.

The logic levels of the digital inputs/outputs are dependent on the I/O bank that they are on. DIO5 to DIO9 are powered by VDDO1 (connected to VDBL inside the hybrid); DIO20 to DIO29 are powered by VDDO3 (the powering of this bank is configurable).

The DIO signal lines from Ezairo 7150 SL provide access to a wide variety of interfaces, including:

- GPIO (controlled by the CFX DSP or controlled by the ARM® Cortex®-M3 processor)
- SPI (x2)
- I^2C
- UART
- LSAD inputs (x6)
- Output Clocks
- Other digital inputs and outputs

For more information about the DIOs, refer to the *Hardware Reference Manual for Ezairo 7100*.

The motherboard provides access to any of the DIOs or their multiplexed signals via the DIO header.

The digital I/O signals on Ezairo 7100 are multiplexed with several interfaces. Refer to the *Hardware Reference Manual for Ezairo 7100* for information regarding the DIO multiplexed signals.

4.5 I²C

The Promira Serial Platform, CAA, or HI-PRO is used to communicate with Ezairo 7150 SL using an I²C communication protocol. The 6-pin DIN header (J2) on the motherboard connects Ezairo 7150 SL to the programmer.

The I²C connection carries I²C bus signals, as well as power, ground, and the VDDO3 signal. Table 7 shows the pin number and associated signal of the J2 connector.

Table 7. DIN Connector (J2) Pin-Out on the Motherboard

Pin Number	J2 Pin Connector
1	Supply Voltage from programmer
2	System Ground
3	SCL (1²C Clock
4	SDA (I²C data)
5	VDDO3 (Sense Voltage)
6	No Connect

4.6 CS44

The motherboard provides a non-populated footprint for a CS44 connector through the VBAT-TP, GND-TP, SCL-TP, and SDA-TP pads.

4.7 Audio Interfaces

4.7.1 Input Stage

The audio input interface consists of the following connectors (standard 0.1" headers) located on the motherboard:

- AI0H
- AI1H
- AI2H
- AI3H

The input ground reference for the above input connectors is normally referenced internally to GND_MIC. The board provides two headers to change the input ground reference to either GND_MIC or to GNDA. Header MICGND configures the input ground reference.

For electrical testing of Ezairo 7150 SL, we recommend configuring the input ground to GNDA when the test equipment provides a single ended (unbalanced) output for supplying the audio input signal. For test equipment that provides differential outputs (balanced), you can connect the input ground to GND_MIC. We recommend performing electro-acoustic testing (microphone and receiver attached to the PCB) with the input ground connected to GND_MIC.

4.7.2 Output Stage

There are two digital outputs (RCVR0, RCVR1) provided on the motherboard. The audio output interface consists of the following connectors located on the motherboard:

- Standard 0.1” headers: FILTEN0, FILTEN1, OUT0 and OUT1
- Two RC filter networks for use with the direct digital outputs when running electrical measurements.

Outputs for the direct digital outputs are available on headers FILTEN0 and FILTEN1. Separate RC filter networks are provided to attenuate out-of-band noise from the direct digital outputs when connected to high impedance audio measurement equipment. The on-board RC filters are enabled by the FILTEN0 and FILTEN1 headers as described in Table 8.

Table 8. On-board RC Filter Configuration

Function	FILTEN0	FILTEN1
Disabled	Open pins 1 to 2 Open pins 3 to 4	Open pins 1 to 2 Open pins 3 to 4
Enabled	Short pins 1 to 2 Short pins 3 to 4	Short pins 1 to 2 Short pins 3 to 4

To connect receivers or loud speakers (without the RC filter) use headers FILTEN0 and FILTEN1 pins 1 and 3.

4.8 Board Configurations

The following section describes the configuration required for powering the different evaluation board setups.

4.8.1 Motherboard and Daughterboard

If you are using both the motherboard and daughterboard, the following subsections describe how to configure them for power.

Daughterboard Configuration

To identify the headers on the daughterboard, see Figure 9 on page 17.

NOTE: Remove solder bridges SB1 and SB2 if they are shorted.

The rest of the configurations depend on the power supply input(s) that you choose. See Table 9.

Table 9. Power Configuration Options for the Daughterboard when Used with the Motherboard

Power Supply Input	Source		
	VDBL	Motherboard Configuration	Other
RFVDD	Populate R4 on the daughterboard with an 11.8 Ω resistor and remove R5. (Default)	Remove R4 and R5 on the daughterboard	**External Supply:** Populate R5 on the daughterboard with an 11.8 Ω resistor and remove R4.
VDDO2	Populate R8 on the daughterboard with a 0 Ω resistor and remove R9. (Default)	Remove R8 and R9 on the daughterboard	**VBAT:** Populate R9 on the daughterboard with a 0 Ω resistor and remove R8.
VDDO3	Populate R7 on the daughterboard with a 0 Ω resistor and remove R3	Remove R3 and R7 on the daughterboard	**VBAT:** Populate R3 on the daughterboard with a 0 Ω resistor and remove R5. (Default)

Motherboard Configuration

To identify the headers on the motherboard, see Figure 8 on page 16. Do the following:

- Add a jumper to the PSU-SEL header to indicate the powering method for Ezairo 7150 SL.
- Connect a jumper or ammeter to the VBAT-I header.
- Set SW3 to ON to power the system.

IMPORTANT: Do not use a multimeter to measure current on the motherboard because the impedance is unsuitable and results in a non-functioning circuit.

The rest of the configurations depend on the resistors that are populated on the daughterboard:

- The RFVDD-SEL header can only be configured with a jumper if the R4 and R5 resistors on the daughterboard are not populated.
- The VDDO2-SEL header can only be configured with a jumper if the R8 and R9 resistors on the daughterboard are not populated.
- The VDDO3-SEL header can only be configured with a jumper if the R3 and R7 resistors on the daughterboard are not populated.

4.8.2 Daughterboard Only

To identify the headers on the daughterboard, see Figure 9 on page 17. For all configurations:

- Short the SB1 and SB2 solder bridges on the daughterboard.
- Connect an external power source to the VBAT/GND pins to power the system.

The rest of the configurations depend on the power supply input(s) that you choose. See Table 10.

Table 10. Power Configuration Options for the Daughterboard when Used Alone

Power Supply Input	Source		
	VDBL	**EXT-RFVDD (external)**	**VBAT**
RFVDD	Populate R4 with an 11.8 Ω resistor and remove R5. (Default)	Populate R5 with an 11.8 Ω resistor and remove R4.	N/A
VDDO2	Populate R8 with a 0 Ω resistor and remove R9. (Default)	N/A	Populate R9 with a 0 Ω resistor and remove R8.
VDDO3	Populate R7 with a 0 Ω resistor and remove R3.	N/A	Populate R3 with a 0 Ω resistor and remove R7. (Default)

APPENDIX A

Connectors

A.1 Overview

This appendix contains a listing of all connectors on the hybrid demonstrator boards. It is divided into the following sections:

- Jumpers: a complete listing and description of all jumpers and possible configurations
- Headers: a complete listing and description of all headers
- Switches: a complete listing and description of all switches and possible configurations
- Connectors: a complete listing and description of all connectors

A.2 Jumpers

Table 11. Motherboard Jumper Descriptions

Designator	Description
RFVDD-SEL	RFVDD (nRF51822 supply) voltage selection
RVDD-I	RFVDD supply current header and power enable
CAA-GND	Disconnects the I^2C programmer (e.g., Promira, Hi-Pro, or CAA) ground from the system ground
VDDO2-SEL	Ezairo 7150 SL VDDO2 voltage selection
VDDO3-SEL	Ezairo 7150 SL VDDO3 voltage selection
VBAT-I	Ezairo 7150 SL VBAT current header and power enable
PSU-SEL	Ezairo 7150 SL VBAT voltage selection
MICGND	Input ground connection to AGND or GNDMIC
FILTEN0	Enable OD0 on-board RC output filter
FILTEN1	Enable OD1 on-board RC output filter
X2	Header for connecting VDDO3 to the CAA VSENSE signal

A.3 Headers

Table 12. Motherboard Header Descriptions

Designator	Description
RFVDD	RFVDD measurement header
RFVDD-EXT	External power supply header for RFVDD
VDDO2-EXT	External power supply header for VDDO2
VDDO3-EXT	External power supply header for VDDO3
DIO	DIO signals for DIOs: 5, 6, 8, 9, 20, 21, 22, 23, 24, and 29
EXT-PSU	External power supply header for VBAT
X3	Header for optional battery capacitor
VBAT	VBAT measurement header
AI0	AI0 input header
AI1	AI1 input header
AI2	AI2 input header
AI3	AI3 input header

Table 12. Motherboard Header Descriptions (Continued)

Designator	Description
OUT0	OD0 filtered output
OUT1	OD1 filtered output
FILTEN0	OD0 header, non-filtered output at pins 1 and 3
FILTEN1	OD0 header, non-filtered output at pins 1 and 3

A.4 SWITCHES

Table 13. Switch Descriptions

Designator	Description	Board
PSU-SEL	Main power switch select	Motherboard
SW1	Momentary push-button switch for selected DIO	Motherboard
SW2	Momentary push-button select for selected DIO	Motherboard
SW3	Main power (VBAT) ON/OFF switch	Motherboard

A.5 CONNECTORS

Table 14. Motherboard Connector Descriptions

Designator	Description
J1	Reserved for ON Semiconductor internal use
J2	6-pin DIN connector to any I^2C programmer, such as Promira, or ~~Hi-Pro~~HI-PRO, or other Ezairo 7100 development board
X1	Connector header for the daughterboard

Table 15. Daughterboard Connector Descriptions

Designator	Description
X1	Connect header for the motherboard

APPENDIX B

Motherboard Schematics

B.1 OVERVIEW

This appendix contains all schematics for the hybrid demonstrator board motherboard:

- Daughterboard Connector
- Communication Interfaces, Main PSU
- Input/Output Connector

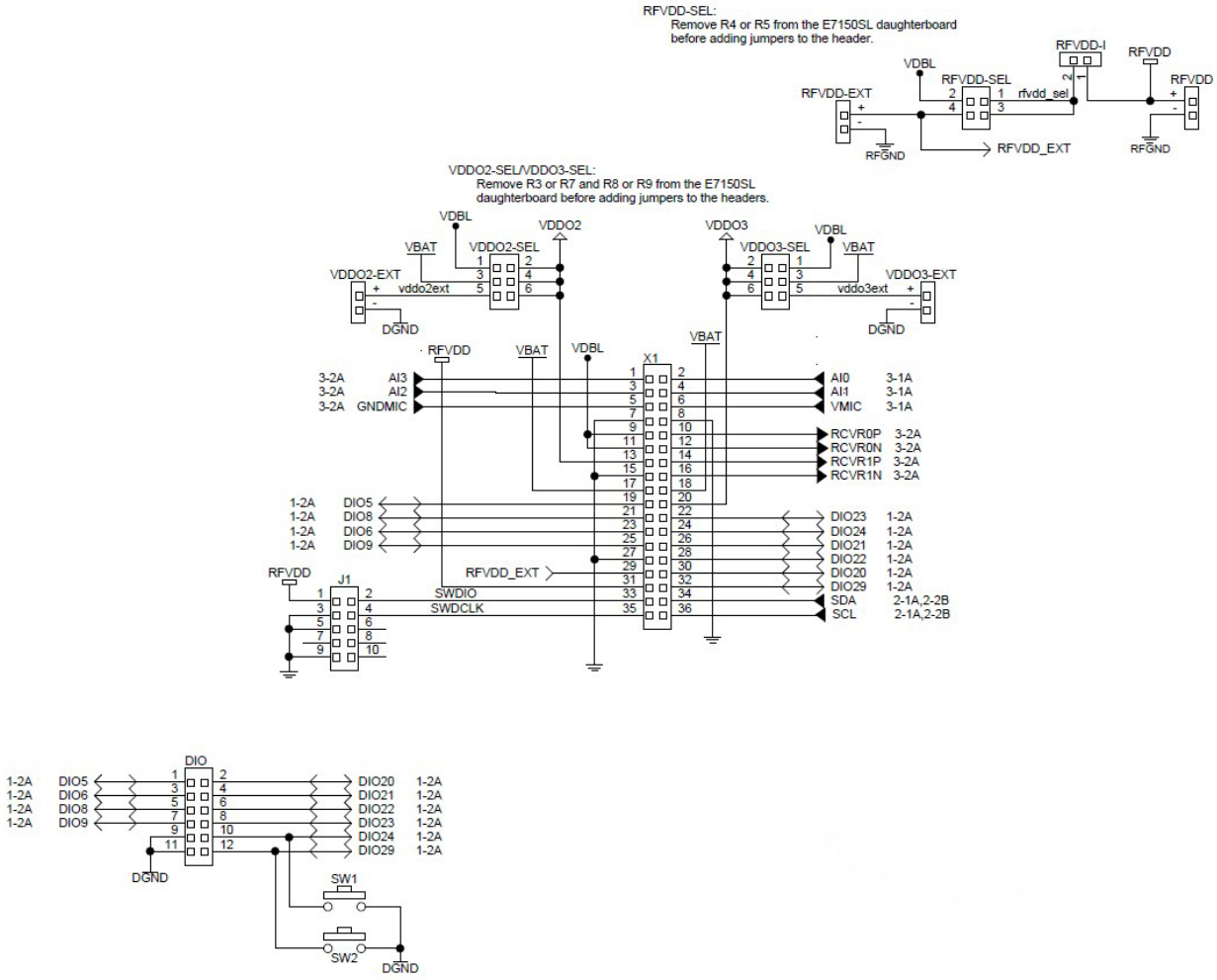


Figure 11. Motherboard: Daughterboard Connector

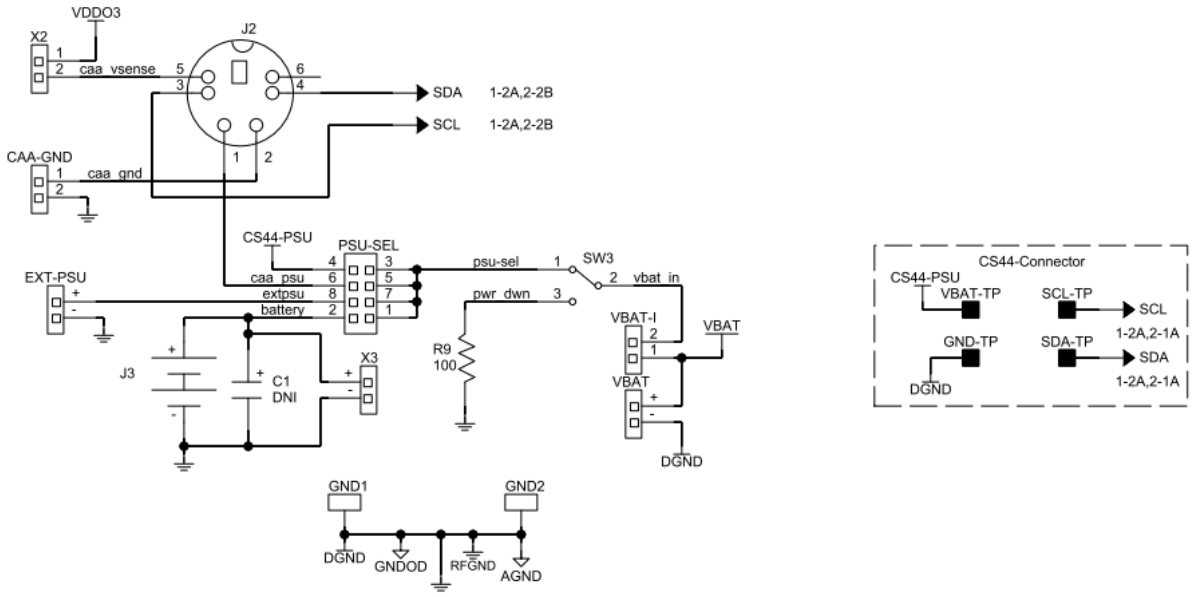


Figure 12. Motherboard: Communication Interfaces, Main PSU

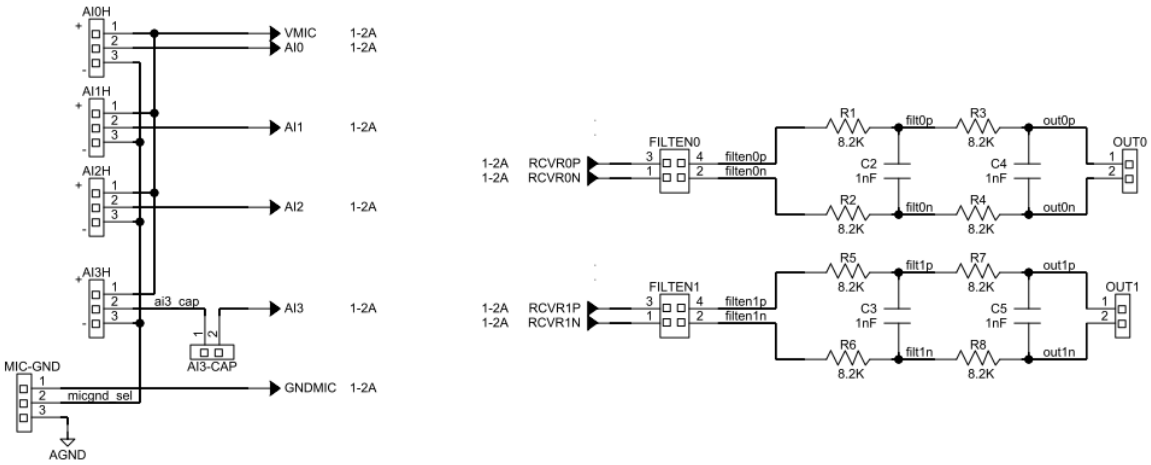


Figure 13. Motherboard: Input/Output Connectors

APPENDIX C

Daughterboard Schematics

C.1 OVERVIEW

This appendix contains the schematic for the hybrid demonstrator board single-ended antenna version of the daughterboard.

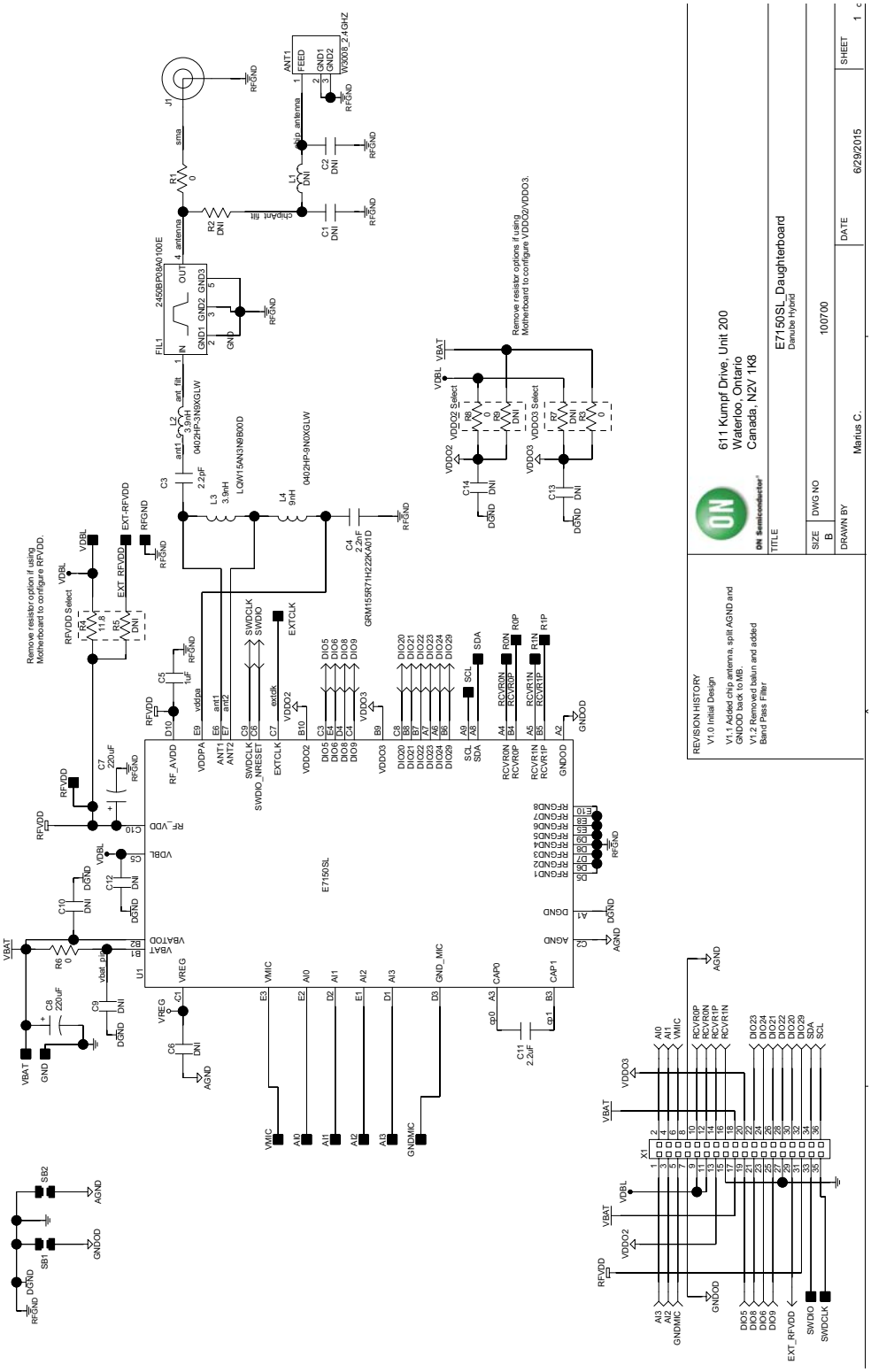


Figure 14. Daughterboard: Ezairo 7150 SL, Motherboard Connector

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